Machine Learning 1
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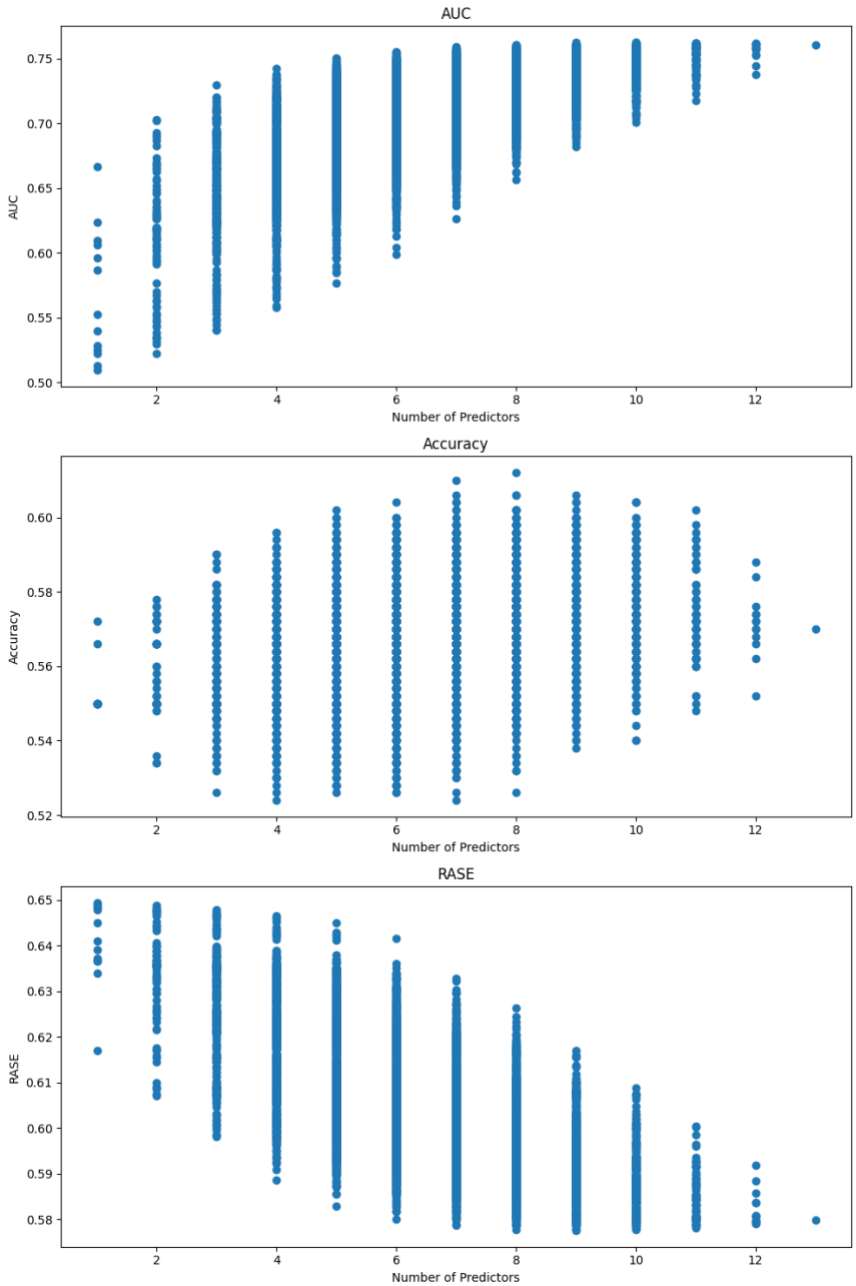
Due 3/11/26

Assignment 4

Question 1 (25 points)
Please consider plotting these three graphs together in a single chart, ideally arranged in a 3 x 1 layout:
the overall classification accuracy versus the number of predictors in a subset, the Root Average Squared Error (RASE) versus the number of predictors, and the Area Under the Curve (AUC) versus the number of predictors. What obvious insights might you discover from these graphs?



For AUC, as the number of predictors increases, we see a positive linear trend. On the contrary, RASE decreases as the number of predictors increases, which makes sense because as the model becomes more accurate and classifies more observations correctly, the error of the model will decrease. Accuracy follows a slightly increasing linear trend, but is more variable and scattered.

Question 2 (25 points)

Could you please provide a table listing the optimal predictor subset for each of the three metrics, along with the corresponding values for all three? It would be helpful to see which predictor subsets perform best for each criterion. By the way, what predictors are included in all three top subsets?

	Metric	Accuracy	AUC	RASE	Best Predictors
0	AUC	0.574	0.762569	0.577704	(av3, chf, miord, mitype, sho, age_bin, bmi_bin, diasbp_bin, hr_bin, sysbp_bin)
1	Accuracy	0.612	0.756297	0.578752	(av3, chf, miord, sho, age_bin, bmi_bin, diasbp_bin, hr_bin)
2	RASE	0.568	0.762504	0.577600	(av3, chf, mitype, sho, age_bin, bmi_bin, diasbp_bin, hr_bin, sysbp_bin)

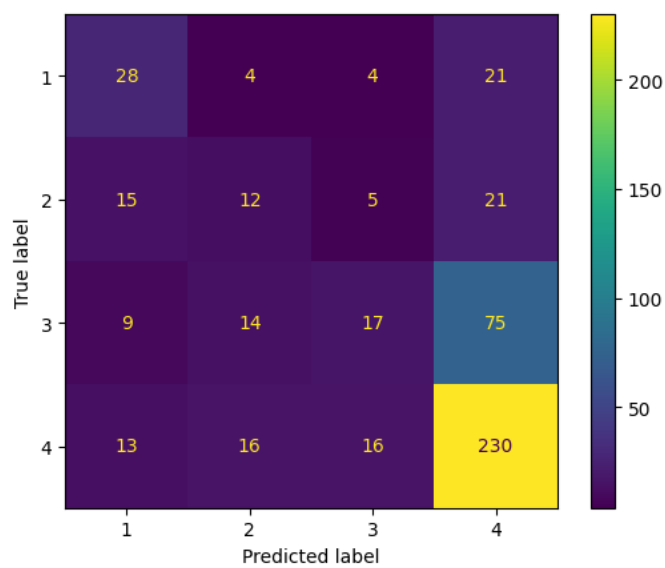
Predictors in all 3 subsets: {'av3', 'chf', 'bmi_bin', 'sho', 'age_bin', 'hr_bin', 'diasbp_bin'}

Question 3 (25 points)

Please share the classification accuracy for each label category along with the Confusion Matrices of your top-performing models in Question 2.

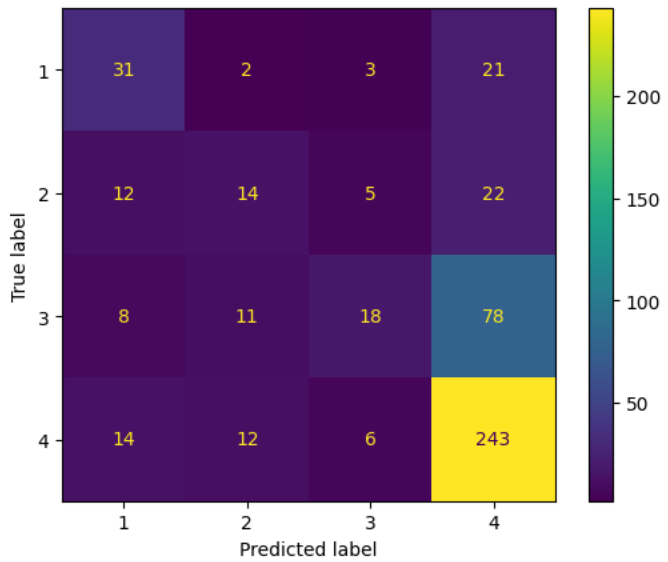
AUC:

	precision	recall	f1-score	support
1	0.43	0.49	0.46	57
2	0.26	0.23	0.24	53
3	0.40	0.15	0.22	115
4	0.66	0.84	0.74	275
accuracy			0.57	500
macro avg	0.44	0.43	0.41	500
weighted avg	0.53	0.57	0.53	500



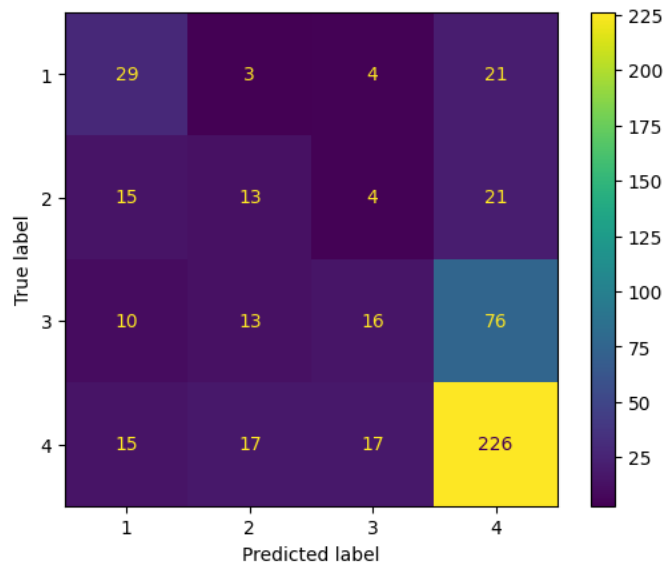
Accuracy:

	precision	recall	f1-score	support
1	0.48	0.54	0.51	57
2	0.36	0.26	0.30	53
3	0.56	0.16	0.24	115
4	0.67	0.88	0.76	275
accuracy			0.61	500
macro avg	0.52	0.46	0.45	500
weighted avg	0.59	0.61	0.56	500



RASE:

	precision	recall	f1-score	support
1	0.42	0.51	0.46	57
2	0.28	0.25	0.26	53
3	0.39	0.14	0.21	115
4	0.66	0.82	0.73	275
accuracy			0.57	500
macro avg	0.44	0.43	0.41	500
weighted avg	0.53	0.57	0.53	500



Question 4 (25 points)

Based on your previous findings, which model will you ultimately recommend? Your objective is to achieve acceptable classification accuracy across all label categories without compromising the model metrics.

I would choose the accuracy model not only because it has the highest accuracy rate, but the model's AUC and RASE are still very close to the respective best models. Additionally, while all models struggle with an imbalance of recall among classes, the model has the best balance of precision among classes, so the model is the best at predicting all classes, not just certain classes. Finally, the model has the lowest amount of predictors making it more efficient and less likely to overfit.